

Database

A database is an organized collection of structured information, or data, stored electronically in a computer system.

It allows users and applications to efficiently store, retrieve, update, and manage vast amounts of information.

How it Works

A database does not just hold files; it categorizes data so it can be quickly searched and filtered. It is typically controlled by a **Database Management System (DBMS)**, which is the software used to interact with the data, handle security, and ensure data integrity.

Importance of database

- Data Storage (A Safe Home)
 - A database provides a centralized and organized place to store information.
 - It acts like a well-structured filing cabinet for digital data.
- Data Retrieval (Fast Finding)
 - Databases allow users to quickly search and access specific information using queries.

- Data can be retrieved in seconds, even from very large collections.
- Data Accuracy (Reduced Errors)
 - Built-in rules and validations help prevent incorrect data entry.
 - This ensures information remains reliable and consistent.
- Data Security (Protection of Information)
 - Databases use authentication, permissions, and access controls.
 - Only authorized users can view, modify, or delete sensitive data.
- Backup and Recovery (Safety Net)
 - Regular backups protect data from accidental loss, hardware failure, or system crashes.
 - Lost information can be restored when needed.
- Time Efficiency (Increased Productivity)
 - Organized storage and fast retrieval save significant time.
 - Routine tasks and reports can be completed much more quickly.
- Data Sharing (Better Collaboration)
 - Multiple users can access and update the same database simultaneously.

- Changes become available to everyone in real time, improving teamwork.

Applications of a Database

Databases are used in almost every field to store, manage, and retrieve information efficiently. Some common applications include:

Banking Systems

- Store customer accounts, transaction records, loan details, and ATM operations.
- Enable secure and fast financial transactions.

Education Institutions

- Manage student records, admissions, examination results, attendance, and library systems.
- Help schools and colleges organize academic information.

Healthcare and Hospitals

- Store patient records, medical histories, appointments, prescriptions, and billing information.
- Improve patient care and data accessibility.

E-Commerce and Online Shopping

- Manage product inventories, customer information, orders, payments, and deliveries.
- Support smooth online business operations.

Library Management Systems

- Keep records of books, members, borrowing, and returns.
- Simplify cataloging and book tracking.

Telecommunication

- Store customer details, call records, billing information, and service plans.
- Help telecom companies manage millions of users efficiently.

Government Organizations

- Maintain records such as citizenship data, tax information, voter registration, and public services.
- Ensure efficient administration and record keeping.

Airline and Railway Reservation Systems

- Manage schedules, ticket bookings, passenger information, and seat availability.
- Provide real-time reservation services.

Business and Human Resource Management

- Store employee records, payroll information, performance evaluations, and attendance.
- Support organizational decision-making.

Social Media Platforms

- Store user profiles, posts, messages, comments, and connections.
- Enable quick access to vast amounts of user-generated content.

Data

Data is a collection of raw facts, figures, observations, or symbols that have not yet been processed or interpreted. By itself, data has little meaning, but when it is processed and organized, it becomes **information**.

Data refers to unprocessed facts and figures that are collected for reference or analysis.

Examples of Data

- Student names
- Roll numbers
- Marks
- Ages
- Dates of birth
- Phone numbers
- Product prices
- For example:
- **85, Ram, 2026-06-29, 9841234567** are all examples of data.

Information

Information is data that has been processed, organized, and interpreted to make it meaningful and useful. It helps people understand situations, make decisions, and solve problems.

Definition

Information is the processed and organized form of data that has meaning and is useful for decision-making.

Examples of Information

- The average marks of a class are **78%**.
- Ram Sharma secured the **highest marks** in the examination.
- Total monthly sales increased by **15%**.
- There are **250 students** enrolled in the BBS program.

DBMS(Database Management System)

A **Database Management System (DBMS)** is software that allows users to create, store, organize, retrieve, update, and manage data in a database efficiently and securely. It acts as an interface between the database and the users or application programs.

A Database Management System (DBMS) is software that enables users to create, access, manipulate, and manage databases while ensuring data security, integrity, and consistency.

Functions of a DBMS

- **Data Storage:** Stores data in an organized manner.
- **Data Retrieval:** Retrieves data quickly using queries.
- **Data Insertion:** Adds new records to the database.
- **Data Update:** Modifies existing records.
- **Data Deletion:** Removes unwanted records.
- **Data Security:** Protects data from unauthorized access.
- **Backup and Recovery:** Recovers data in case of system failure.
- **Data Integrity:** Maintains the accuracy and consistency of data.
- **Concurrency Control:** Allows multiple users to access the database simultaneously without conflicts.

Advantages of DBMS

- Reduces data redundancy (duplicate data).
- Improves data consistency.
- Enhances data security.
- Enables data sharing among multiple users.
- Provides backup and recovery facilities.
- Ensures data integrity.
- Makes data retrieval faster and easier.

- Supports multiple users at the same time.

Examples of DBMS

- MySQL
- Oracle Database
- Microsoft SQL Server
- PostgreSQL
- SQLite
- MariaDB
- IBM Db2
- MongoDB (NoSQL DBMS)

Applications of DBMS

- Banking systems
- Educational institutions
- Hospitals
- Library management systems
- Railway and airline reservation systems
- E-commerce websites
- Telecommunications
- Government organizations

Database Languages

Database languages are specialized languages used in a **Database Management System (DBMS)** to create, manage, retrieve, update, and control data in a database. They provide commands that allow users and database administrators to interact with the database efficiently.

SQL (Structured Query Language)

SQL (Structured Query Language) is the standard language used to communicate with a database. It allows users to create databases, store data, retrieve information, update records, and delete data easily.

Database languages are the **types or categories** of database commands.

SQL is the **language** that provides these commands.

Tables in DBMS

A **table** is the basic structure used to store data in a relational database. It organizes data into **rows** and **columns**, making it easy to store, retrieve, and manage information.

Rows (Records/Tuples): Each row represents one complete record.

Columns (Fields/Attributes): Each column represents a specific type of information about the records.

Example: Student Table

Student_ID	Name	Age	Department
101	Ram	20	BCA
102	Sita	19	BBS
103	Hari	21	BSc

Queries in DBMS

A **query** is a request made to a database to **retrieve, insert, update, or delete data**. Queries are written using **SQL (Structured Query Language)** and allow users to interact with the database.

A query is a request or command used to access or modify data in a database.

Report in DBMS

A **report** is a formatted presentation of data retrieved from a database. It organizes and displays information in a clear and readable format, making it easier to analyze, print, and share.

A report is a structured and formatted output of data from a database used for viewing, printing, and decision-making.

DDL (Data Definition Language)

DDL (Data Definition Language) is a subset of SQL used to define, modify, and manage the structure of database objects such as tables, schemas, indexes, and views.

Common DDL Commands

Command	Purpose
CREATE	Creates a new database object (table, view, index, etc.)
ALTER	Modifies the structure of an existing object
DROP	Deletes an object permanently

Examples

CREATE Table

```
CREATE TABLE Employee (  
    EmpID INT PRIMARY KEY,  
    Name VARCHAR(50) ,  
    Salary DECIMAL(10,2)  
);
```

2.ALTER Table

a) Add a New Column

```
ALTER TABLE Employee  
ADD Department VARCHAR (30) ;
```

b) Modify an Existing Column

Changes the data type or size of an existing column.

MySQL/XAMPP:

```
ALTER TABLE Employee  
MODIFY Name VARCHAR (100) ;
```

C) Delete a Column

```
ALTER TABLE Employee  
Delete Department;
```

3. DROP Table

```
DROP TABLE Employee;
```

DML (Data Manipulation Language)

DML (Data Manipulation Language) is a set of SQL commands used to **insert, retrieve, update, and delete data** in database tables. Unlike DDL, DML works with the **data inside the tables**, not the table structure.

1. INSERT

Adds new records (rows) to a table.

For Single Entry Syntax:

```
INSERT INTO table_name (column1, column2,  
...)  
VALUES (value1, value2, ...);
```

For Multiple Entry Syntax:

```
INSERT INTO table_name (column1, column2, column3)  
VALUES  
(value1, value2, value3),  
(value4, value5, value6),  
(value7, value8, value9);
```

Example:

```
INSERT INTO Employee (EmpID, Name, Salary,  
Department)  
VALUES (101, 'Ram Sharma', 45000.00, 'IT');
```

2. SELECT

Retrieves data from a table.

Syntax:

```
SELECT column1, column2  
FROM table_name;
```

Example (Display all records):

```
SELECT * FROM Employee;
```

Example (Display specific columns):

```
SELECT Name, Salary  
FROM Employee;
```

Example (Display all columns using where and condition):

```
SELECT *  
FROM table_name  
WHERE condition;
```

3. UPDATE

Modifies existing records in a table.

Syntax:

```
UPDATE table_name  
SET column_name = value  
WHERE condition;
```

Example:

```
UPDATE Employee  
SET Salary = 50000.00  
WHERE EmpID = 101;
```

4. DELETE

Removes one or more records from a table.

Syntax:

```
DELETE FROM table_name  
WHERE condition;
```

Example:

```
DELETE FROM Employee  
WHERE EmpID = 101;
```

Note: Omitting the WHERE clause deletes **all records** from the table.

```
DELETE FROM Employee;
```